

Jyot Antani, PhD

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Research Interests: Biophysics, Imaging Technology, Virology, Host-pathogen Interactions

Special Skills: MATLAB, Image Analysis - Biomatter Tracking, CAD Modeling, Microscopy

Experience & Education

Jun 2021 Onwards	Postdoctoral Fellow, joint appointment in the Emonet and Turner Labs, Yale University (US) Research: Biophysics of interactions between bacteriophage and bacteria
Aug 2016- Jun 2021	PhD Student, Lele Lab , Department of Chemical Engineering, Texas A&M University (US) Dissertation title: Sensory Functions of the Bacterial Flagellar Motor
Jul 2012- May 2016	Bachelor of Technology in Chemical Engineering with Honours Indian Institute of Technology Bombay (IIT Bombay), Mumbai (India)

Doctoral Research

Molecular mechanisms of host substrate sensing by bacterial flagellar motors

Determined the effect of mechanical forces on chemotactic output of the bacterial flagellar motor, with phase microscopy, optical tweezers, and TIRF imaging. Developed a biophysical model for the regulation of allosteric interactions by mechanical force.

Motility, chemotaxis, and random spread of *Helicobacter pylori*, the carcinogenic pathogen

Devised a technique to experimentally determine the chemotactic activity of *H. pylori*. Studying the biophysical principles governing chemotactic migration by this bacterial species which employs run-reversal motility – different from traditionally studied *E. coli*.

Design and development of experimental and analytical framework

Designing 3D-printed imaging chambers for specific requirements in microscopy experiments, e.g., perfusion, imaging side-views. Writing MATLAB-based image analysis codes for tracking motile bacteria, mammalian cell-nuclei, and fluorophores.

Technical Skills

Bacterial Culturing Molecular Biology	Perfusion & Thermal Control Experiments Prototyping & 3D-printing	Microscopy: Dark-field, Optical Tweezing, TIRF, FRET, Image Analysis
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Computer Proficiency / Computational Skills

Image Processing & Analysis:	MATLAB, Python, ImageJ – {Highlight: particle-tracking algorithms}
Design & Development:	SolidWorks, AutoCAD, LabVIEW, Molecular Biology {SnapGene, Benchling}
Simulation Packages:	ANSYS Fluent, Aspen Plus, OpenFOAM, COMSOL, CHARMM, VMD
Programming:	MATLAB, Python, C, C++, Fortran, HTML

Patent

Vertical Substrate Mount for Z-axis Imaging. U.S. Provisional Pat. Ser. No. 63/040,800, filed Jun 18, 2020.

Publications

Antani JD, Sumali AX, Lele TP, Lele PP (2021) Asymmetric random walks reveal that the chemotaxis network modulates flagellar rotational bias in *Helicobacter pylori*. *eLife*. doi:10.7554/eLife.63936

Antani JD, Gupta R, Lee AH, Mandon MD, Lele PP. Proprioception in the *Escherichia coli* flagellar motor. *Under review*.

Katiyar A*, Antani JD*, McKee BP, Gupta R, Lele PP, Lele TP (2020) A method for direct imaging of x-z cross-sections of fluorescent samples. *Journal of Microscopy*. doi:10.1111/jmi.12965 (*equal contribution)

- Wong GCL, Antani JD, et al. (2021) Roadmap on emerging concepts in the physical biology of bacterial biofilms: from surface sensing to community formation. *Physical Biology*. doi:10.1088/1478-3975/abdc0e
- Ford KM, Antani JD, Nagarajan A, Johnson MM, Lele PP (2018) Switching and torque generation in swimming cells of *E. coli*. *Frontiers in Microbiology*, 9, 2197. doi:10.3389/fmicb.2018.02197
- Antani JD, Gupta R, Lele PP. The chemotaxis conundrum for a polar-flagellated bacterium. *In preparation*.
- Katiyar A, Zhang J, Antani JD, Yu Y, Lele PP, Reinhart-King CA, Sniadecki N, Roux K, Dickinson R, Lele TP. Liquid like nuclear deformation around extracellular obstacles during cell migration. *In preparation*.

Conference Presentations

Antani JD, Lele PP. Modulation of chemotactic signaling in bacteria by mechanical forces.

- Mar 2019 | ChEGSA Annual Symposium, Texas A&M University, College Station (USA).
 Oct 2018 | American Institute of Chemical Engineers (AIChE) Annual Meeting, Pittsburgh (USA).
 Dec 2017 | Texas Branch of American Society for Microbiology, 2017 Fall Meeting, College Station (USA).

Antani JD, Lele PP. Asymmetric Swimming and Chemotaxis in *Helicobacter pylori*.

- Jan 2021 | Bacterial Locomotion And Signal Transduction (BLAST) XVI Meeting, Zoom (online).
 Feb 2020 | Biophysical Society (BPS) Annual Meeting, San Diego (USA).
 Jan 2019 | Bacterial Locomotion And Signal Transduction (BLAST) XV Meeting, New Orleans (USA).

Antani JD, Venkatesh KV. Analysis of Glucose Homeostasis in Humans.

- Nov 2015 | Futuristic Approach for Alternatives to Animal Testing CEFIPRA seminar, IIT Bombay, India.

Honors & Awards

- 2020 | CIRTLL Associate Certificate, Center for the Integration of Research, Teaching, and Learning
 2015 | Best Poster Award, *Futuristic Approach to Alternatives* seminar, sponsored by Indo-French Centre for the Promotion of Advanced Research (CEFIPRA) and L'Oréal India, location: IIT Bombay
 2015 | Undergraduate Research Award URA01, IIT Bombay
 2008 | National Talent Search Scholarship, NCERT, India

Additional Research Experience

- Jul 2015-
Feb 2016 | Analysis of Glucose Homeostasis in Humans
 Guide: Prof. K. V. Venkatesh, IIT Bombay
 Formulated a Hill function-based empirical biochemical model of body glucose concentration following a fixed food-intake, obtained least-squares fits to data acquired from 150+ individuals. Constructed a classification tree based on fit-parameters using machine learning in MATLAB; established a statistical grouping among Diabetic and Healthy individuals.
- May-Jul
2015 | A Novel Protocol for Coating Lipid Bilayers onto Silver Nanocubes
 Guide: Dr. Hung-Jen Wu, Texas A&M University
 Established a protocol to use a tip-sonicator for making small unilamellar vesicles of lipids, to coat lipid bilayers onto Silver nanocubes. Employed a Localized Surface Plasmon Resonance-based nanocube sensor to monitor protein binding on supported lipid bilayers. Analyzed binding kinetics of GM1 ganglioside with Cholera Toxin Subunit B.
- May 2014-
Apr 2015 | Design of a Chemical Vapor Deposition Reactor
 Guide: Prof. Mahesh S. Tirumkudulu, IIT Bombay
 Simulated fluid flow around a disk, successfully obtained the pressure coefficients observed experimentally in literature. Proposed an unconventional design for uniform coating of parylene polymer onto silicon micro-wafers in a micro-scale Chemical Vapor Deposition reactor. Modeled Parylene flow using ANSYS Fluent, enhanced the proposed design.
- Jun 2013 | Process Improvement and Technology in Jaggery-Making
 Guide: Prof. Sanjay M. Mahajani, IIT Bombay
 Examined units under development for continuous production as well as recuperation in manufacturing Jaggery, the traditional non-centrifugal cane sugar. Analyzed Jaggery samples for acidity, moisture content, water activity, density, sugar content etc; observed the XRD and ESEM techniques.

Teaching & Mentoring Experience

- Fall 2019 | A hands-on workshop for image processing and single-particle tracking in MATLAB
Pushkar Lele Lab, Texas A&M University
- Apr 2017-
Apr 2018 | AggieE_Challenge – Undergraduate Mentoring, Texas A&M University
Graduate-mentor to a multidisciplinary team of eight undergraduate students employing 3D-printing technology to design scaled-up bacterial models, in order to study the hydrodynamic effects of cell morphology on motility.

Teaching Assistant at Texas A&M Chemical Engineering
Heat Transfer Operations (Spring 2019)

Bioprocess Engineering (Spring 2017, Fall 2016)

Teaching & Mentoring Training

- Aug 2016-
Dec 2019 | [Academy for Future Faculty \(AFF\)](#), Center for Teaching Excellence, Texas A&M University
Faculty Mentor: Dr. Micah Green, Texas A&M University
A program designed for training graduate students as future faculty members, featuring weekly seminars and workshops, and anchored by faculty mentorship.
- Jul-
Aug 2019 | An Introduction to Evidence-Based Undergraduate STEM Teaching
Muti-university Massive Open Online Course from the [NSF-founded CIRTl network](#)
Eight-week online course on [edX.org](#) catering introduction to effective teaching strategies and the research that supports them. Attended local learning community meetings for the course at Texas A&M University.
- Oct 2019 | Improving Research Mentoring Relationships – Introduction to Effective Mentorship
Center for the Improvement of Mentored Experiences in Research (CIMER)
1-day workshop at Texas A&M University, conducted by CIMER from University of Wisconsin-Madison.

Leadership Experience

- Oct 2018-
Aug 2020 | Secretary and Chair, Chemical Engineering Graduate Student Advisory Council
Artie McFerrin Department of Chemical Engineering, Texas A&M University
Addressed graduate student welfare through communication and feedback loops with department administration.
- Apr 2014-
Mar 2015 | Manager, Competitions, AZeotropy 2015
IIT Bombay's Intercollegiate Chemical Engineering Symposium
Pioneered a new model-making competition *ChemETimer* based on time-precision. Designed a presentation-based competition *Swaasthya* on engineering-applications to healthcare. Designed *Cipher*, the online cryptic hunt which received a participation of 400+. Actualized unconventional chemical engineering quizzing for *Chem-O-Philia*, national level quiz.

Key Course Projects

- Spring 2018 | Determining Diffusion Coefficient of Microparticles using Particle-Tracking Algorithms
Course: Colloidal Applications and Biomolecular Interactions in Life Science
Employed particle-tracking algorithms in MATLAB to track the Brownian diffusion of submicron-sized particles recorded using phase-contrast microscopy. Determined the diffusion coefficient of the particles by calculating the average mean-squared displacement of an ensemble.
- Spring 2017 | Project Proposal: Evolution of Bacterial Flagellum *Course: Evolutionary Bioinformatics*
Proposed a study involving the comparison of flagellar genome across multiple bacterial species which has potential to aid in obstruction of pathogenic motility. Carried out BLAST alignment analysis for nucleotide and protein sequences of flagellin (flagellar filament) gene as preliminary results.
- Spring 2016 | Review: How Deep a Cell Can Sense *Course: Cell Mechanics*
Carried out a literature survey and reported theoretical, in-vitro as well as in-silico studies on quantified investigations about the range of mechanical signaling.